

DLOps Assignment-5 Report

LoRA on ViT-S and IBM ART Adversarial Experiments

April 2026

1 Execution Environment

The Q1 experiments reported here were executed in a CPU-only workflow for training, evaluation, attack generation, detector training, plotting, and model export. The Q2 section below has been updated using the latest recorded run log in the workspace.

2 Q1: LoRA on ViT-S for CIFAR-100

2.1 Objective

The goal was to finetune an ImageNet-pretrained ViT-S model on CIFAR-100 in two settings:

- baseline head-only finetuning without LoRA,
- LoRA-based adaptation over multiple rank and alpha combinations with dropout fixed at 0.1.

The assignment also required train/validation metrics, test comparison tables, class-wise accuracy histograms, gradient-update plots on LoRA weights, Optuna tuning, and model publication.

2.2 Implementation Summary

The final Q1 pipeline uses the following design choices:

- Base model: `vit_small_patch16_224` from `timm`, pretrained on ImageNet.
- Dataset: CIFAR-100.
- Baseline: backbone frozen, classification head trainable.
- LoRA grid: ranks $\{2,4,8\}$, alpha $\{2,4,8\}$, dropout = 0.1.
- Data preprocessing: resize to 224×224 , CIFAR-100 normalization, random crop, and horizontal flip.
- Optimizer: AdamW.
- Scheduler: cosine annealing.
- Additional logging: LoRA gradient norms were recorded and plotted.

2.3 Important Note on Q, K, V Injection

The assignment asks for LoRA injection into the attention weights of Q , K , and V . In the `timm` ViT implementation used here, these projections are exposed as a **single fused** linear layer named `qkv` rather than separate `q_proj`, `k_proj`, and `v_proj` modules. Therefore, LoRA was applied to the fused `qkv` projection. This is the correct implementation choice for this codebase and is explicitly documented here to avoid ambiguity.

2.4 Recorded Training Metrics

The saved CSV histories in the workspace contain one recorded epoch entry for each experiment. Table 1 consolidates those saved values.

Table 1: Q1 recorded training summary from saved history CSV files.

Experiment	Train Loss	Val Loss	Train Acc	Val Acc	LoRA Grad Norm
baseline	1.2034	0.7585	0.7155	0.7887	0.0000
lora_r2_a2	0.8025	0.3995	0.8003	0.8764	25452.0160
lora_r2_a4	0.7781	0.3910	0.8065	0.8768	37978.1629
lora_r2_a8	0.7756	0.4019	0.8052	0.8840	53662.8436
lora_r4_a2	0.8055	0.3989	0.7999	0.8800	18169.9222
lora_r4_a4	0.7918	0.4048	0.8003	0.8806	27059.0788
lora_r4_a8	0.7568	0.3939	0.8099	0.8782	38369.1075
lora_r8_a2	0.8070	0.3981	0.8000	0.8796	13571.7572
lora_r8_a4	0.7875	0.3938	0.8047	0.8798	19525.9257
lora_r8_a8	0.7602	0.3812	0.8096	0.8852	27844.0375
lora_partial_freeze	0.7631	0.3836	0.8069	0.8782	27825.9369
optuna_best_r2_a2	0.6246	0.4760	0.8236	0.8576	28270.8415

2.5 Overall Test Results

Table 2 follows the format requested in the assignment and compares baseline and LoRA runs.

Table 2: Q1 overall testing results.

LoRA	Rank	Alpha	Dropout	Test Accuracy	Trainable Params
No	0	0	0.0	0.7791	38500
Yes	2	2	0.1	0.8734	75364
Yes	2	4	0.1	0.8772	75364
Yes	2	8	0.1	0.8746	75364
Yes	4	2	0.1	0.8727	112228
Yes	4	4	0.1	0.8764	112228
Yes	4	8	0.1	0.8778	112228
Yes	8	2	0.1	0.8725	185956
Yes	8	4	0.1	0.8751	185956
Yes	8	8	0.1	0.8775	185956
Yes	8	8	0.1	0.8712	185956
Yes	2	2	0.1	0.8510	75364

2.6 Observations

- The baseline head-only finetuning reached a test accuracy of 0.7791.

- The best fixed-grid LoRA setting was `rank=4`, `alpha=8`, `dropout=0.1`, with test accuracy 0.8778.
- This is an absolute improvement of 0.0987 over the baseline.
- Several LoRA settings performed very similarly, which indicates that the method is robust over a modest hyperparameter range.
- The optional partial-freeze + LoRA run (0.8712) did not improve over the best fully frozen-backbone LoRA result.
- The Optuna-selected configuration trained in the final run achieved 0.8510, which was below the best manually enumerated grid result in this experiment set.

2.7 Optuna and Model Export

Optuna was used to search LoRA-only hyperparameters. The best trial used:

- `rank = 2`,
- `alpha = 2`,
- `learning rate = 0.002718177657750734`.

The best model artifacts from the experiment set were exported and uploaded to Hugging Face Hub under `subilal/lora-vit-cifar100`. WandB logging was also enabled in the script.

2.8 Experiment Links

- Hugging Face model: [subilal/lora-vit-cifar100](#)
- WandB Q1 project: [vit-lora-cifar100](#)
- WandB Q2 project: [dlops-assignment5-q2-art](#)

2.9 Generated Q1 Figures

The best exported run available in the workspace is `lora_r4_a8`. The report includes the available plots for that run when the files are present.

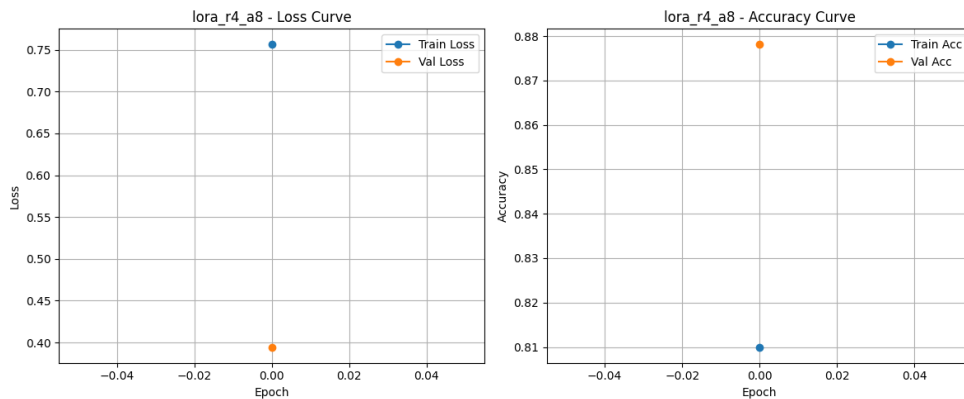


Figure 1: Best Q1 run: train/validation loss and accuracy curves.

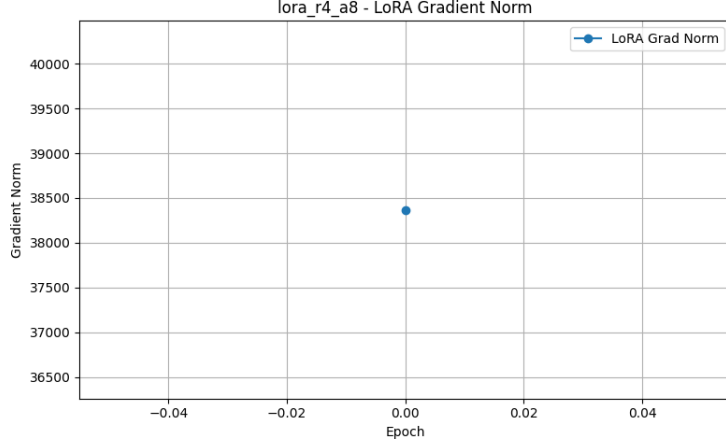


Figure 2: Best Q1 run: LoRA gradient-update graph.

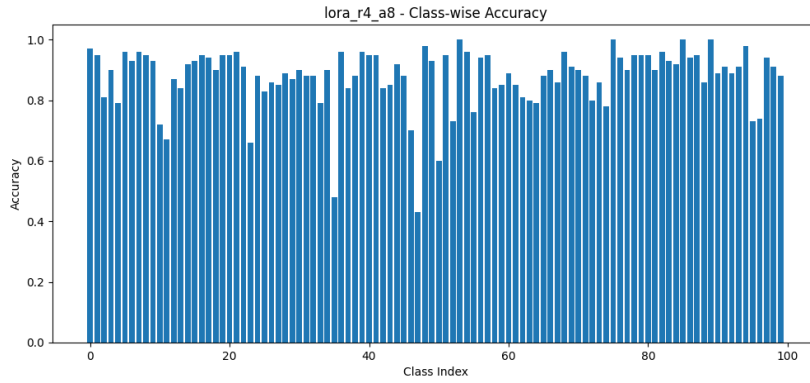


Figure 3: Best Q1 run: class-wise test accuracy histogram.

3 Q2: Adversarial Attacks Using IBM ART

3.1 Objective

This part has two components:

- compare FGSM implemented from scratch versus FGSM from IBM ART on CIFAR-10 using a ResNet-18 classifier trained from scratch,
- train adversarial detectors on mixed clean/adversarial samples for PGD and BIM attacks using ResNet-34 based models.

3.2 Implementation Summary

The Q2 pipeline trains a clean ResNet-18 classifier, evaluates FGSM attacks over multiple epsilon values, and then trains two binary detectors:

- PGD detector: clean vs PGD-adversarial,
- BIM detector: clean vs BIM-adversarial.

The script generates visual comparisons for FGSM scratch, FGSM ART, PGD, and BIM, and logs those artifacts to WandB together with image tables, metrics, and attack summaries. The

current workspace snapshot contains an updated Q2 run log and the corresponding WandB links.

3.3 Clean Classification Accuracy

In the latest recorded run, the clean CIFAR-10 classifier reached a final test accuracy of **0.8350**, which satisfies the assignment requirement of at least 0.72.

3.4 FGSM Comparison

Table 3: FGSM comparison on CIFAR-10 using the latest recorded Q2 run.

Epsilon	Clean Acc	Scratch Acc	ART Acc	Scratch Drop	ART Drop
0.0000	0.8350	0.8350	0.8350	0.0000	0.0000
0.0100	0.8350	0.6971	0.7646	0.1379	0.0704
0.0200	0.8350	0.5568	0.6449	0.2782	0.1901
0.0300	0.8350	0.4302	0.5226	0.4048	0.3124
0.0500	0.8350	0.2517	0.3474	0.5833	0.4876

3.5 Adversarial Detector Results

Table 4: Best detector test accuracy for each attack family.

Detector Attack	Best Detection Accuracy
PGD	0.7147
BIM	0.7215

3.6 Observations

- The clean classifier exceeded the required clean accuracy threshold comfortably.
- Increasing epsilon caused a steady degradation in accuracy for both FGSM implementations.
- In this run, IBM ART FGSM consistently retained higher classifier accuracy than the scratch FGSM implementation at the same epsilon values.
- Both detector models crossed the required 70% detection accuracy threshold.
- The BIM detector was again slightly stronger than the PGD detector in the latest recorded run.
- The updated WandB run logged the attack galleries, comparison plots, tabular summaries, and clean/adversarial sample tables required by the assignment prompt.

3.7 Q2 WandB References

- WandB project: [dllops-assignment5-q2-art](#)
- Latest recorded run link: [worthy-sponge-1](#)

4 Conclusion

This assignment demonstrates two complementary themes: parameter-efficient finetuning and robustness evaluation.

- For Q1, LoRA substantially improved CIFAR-100 performance over baseline head-only finetuning, with the best result coming from `rank=4, alpha=8`.
- For Q2, the clean ResNet-18 classifier achieved strong performance, while adversarial robustness degraded with stronger FGSM perturbations as expected.
- The PGD and BIM detectors both met the minimum performance target.

5 External Links

- Hugging Face model: [subilal/lora-vit-cifar100](#)
- WandB Q1: [vit-lora-cifar100](#)
- WandB Q2: [dlops-assignment5-q2-art](#)